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Examining the impact of flipped classroom learning (FCL) and gamified-flipped classroom learning (G-FCL) on students' attitudes and emotions toward learning in a university course

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Abstract

The flipped classroom, as an active learning methodology compared to the traditional classroom, has a significant impact on students' learning experiences in higher education. However, its positive effects largely depend on students' involvement, motivation, and personal characteristics. This research examines and compares the efficacy of flipped classroom learning (FCL) and gamified-flipped classroom learning (G-FCL), regarding students' affective domain (attitude and emotion), in order to address challenges associated with active learning. The study, conducted over four academic years (2019/20 to 2022/23), used a quasi-experimental design and a quantitative survey of 242 pre-service teachers (PSTs). Results showed that PSTs in the G-FCL group significantly increased their positive attitudes toward learning compared to those in the FCL group. Additionally, students' grades before the study (pre-grade) were found to be a relevant predictor of global attitudes. In terms of emotions, both methodologies had a significant influence, and pre-grade was again confirmed as a relevant predictor of global emotions. Multiple regression analysis of global attitudes and emotions, considering the studied variables (instruction methodology, pre-grade, educational background, and gender), indicated that these were significant predictors. However, there was no statistically significant difference related to PSTs' gender. Overall, the findings reiterated that G-FCL was critical in promoting students' affective domain, especially attitude, compared to FCL, although FCL itself had proved effects in previous research. In addition, it can be noted that employing blended and gamified learning methodologies will provide further benefits in the flipped classroom as an active learning.

Keywords

active learning, affective domain, flipped classroom, gamification, higher education, personal characteristics

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Introduction

Currently, active learning in higher education is highly encouraged due to its potential to improve not only the students' learning outcomes but also the affective domain associated with the learning process, compared to traditional lecture-based education (Blanco et al., 2010). Any teaching methodology that aims to stimulate and motivate students can be considered active learning in the broadest sense (Prince, 2004; C. Williams et al., 2022). This approach originates from the theory of constructivism (Michael, 2006). In an active learning setting, students engage with the learning paradigm by participating in tasks and reflecting on their experiences (Bonwell & Eison, 1991; Roberts, 2017). Particularly, in science, technology, engineering, and mathematics (STEM) education, previous research (Freeman et al., 2014; Roach, 2014; Yoon et al., 2020) has found out that these methods create a proper learning environment where students can enhance their cognitive (learning outcome) and affective domain, that includes students' attitudes, beliefs, and emotions toward the learning process (Blanco et al., 2010; McLeod, 1989). In this context, active learning requires more responsibility from students compared to passive teaching methods. In higher education, active learning can also foster better concept formation and increase students' interest (Li et al., 2019), as it focuses more on student-centered activities (Bishop & Verleger, 2013; O'Flaherty & Phillips, 2015), although the reality of active learning is not that simple. Consequently, the role of proper instructional methodology in active learning should be further deliberated in higher education.

Previous research studies have found that the appropriate instruction methodology significantly impact students' learning experiences (Chen & Law, 2016; Jeong & González-Gómez, 2022; Lee, 2011). Flipped classroom learning (FCL), compared to conventional classroom, particularly in challenging STEM courses, can improve students' learning efficacy and enhance teaching involvement (González-Gómez et al., 2022; Lo et al., 2017; Moraros et al., 2015; M. Williams et al., 2018). The flipped classroom is a technology-enhanced pedagogical approach that emphasizes student-centered methodology through in-class and just-in-time activities, while incorporating various computer-based individual materials for out-of-class activities (Bishop & Verleger, 2013; Jeong & González-Gómez, 2021; Nederveld & Berge, 2015). However, the positive effects in the FCL largely rely on the involvement and motivation of students for the materials and activities given inside and outside of classroom (Durrani et al., 2022; Hao, 2016; Kim et al., 2014). In the FCL context, there have been attempts to improve gamification materials for out-of-class activities (Durrani et al., 2022; Huang & Hew, 2018; Lo & Hew, 2020). Since there are no or few studies that have specifically examined the impact of blended methodology through in-class activity with the gamification materials in the FCL (Bassett et al., 2020), this approach could provide individual guidance as adding a literature of theoretical foundation and make class time more valuable. Thus, there is limited literature comparing only the FCL and gamified-FCL (G-FCL).

Gamification utilizes game-like elements designed in a non-game context to make people involved and motivated (Deterding et al., 2011; Zichermann & Cunningham, 2011). Specifically, Prensky (2003) along with other researchers advanced game-based learning and theory by incorporating game-like elements into the educational and learning context (Dicheva et al., 2015). In the past decade, it has gained importance in achieving goals through the improved student engagement and motivation (Liao et al., 2019; Nicholson, 2012; Sailer et al., 2017). Thus, its impact mostly depends on how well goals and challenges are identified. Additionally, it depends on how responsibility and improvement are handled to achieve the given tasks, examining whether the model is valid for reaching the goals and challenges (Alt, 2023; Liao et al., 2019). Although game-based learning has various advantages in engaging and teaching students, it still faces significant challenges in implementation. Particularly, difficulties arise in designing tailored games and

considering technology proficiency, especially if an instructor is not a game developer (Pho & Dinscore, 2015; Sun et al., 2021). With respect to the increase in student involvement and motivation, very little research has been conducted in a FCL setting for STEM education, despite the increasing educational use of gamification (Dichev & Dicheva, 2017). However, it is necessary to investigate how to design and implement this instructional advance to gauge students' affective domain. Pedagogical schemes, with proper theoretical foundations, should be included in the literature on blended-FCL and G-FCL in the context of the affective domain in higher education.

In education, various multidimensional factors influence the success of active instruction methodologies implemented in higher education classrooms (Bandura, 1971; Fredricks et al., 2004; Hascher & Edlinger, 2009). In the context of STEM education, educational research highlights the importance of considering the affective domain of the teaching and learning process (Furinghetti & Morselli, 2009), which includes key descriptors such as attitudes, emotions, and beliefs (McLeod, 1989). Students' experiences during the learning process evoke feelings and emotional reactions that significantly influence their ability to learn (Blanco et al., 2010). The interplay between cognitive and affective domains is a critical outcome of the learning process (Ashkenazi & Velner, 2023; Huang et al., 2019; Pintrich et al., 1993). Here, the influence of affective domain on learning has been theoretically analyzed in prior studies (Bower, 1981; Hascher & Edlinger, 2009). For instance, Schwarz's (1990) "mood as information theory" suggests that mood significantly affects students' learning experiences. However, research in STEM education has predominantly focused on the cognitive aspects of teaching and learning practices, often neglecting the affective domains (Aragón et al., 2024).

Recently, there has been growing interest in understanding how attitudes (Chang et al., 2024; Joensuu-Salo et al., 2024; Wadaani, 2023) and emotions (Bellocchi et al., 2013; Gündüz & Akkoyulu, 2020; Kyriaki Kourti & Potari, 2024; Pekrun et al., 2007) impact the overall teaching and learning process. While blended gamification in the FCL could motivate students to engage with their studies, it is essential to consider participants variables such as grade, educational background, and gender. Furthermore, it is also necessary to examine whether the proposed model can enhance students' affective domain. Such improvements are crucial for helping students achieve their final goals in higher education.

As the existing research comparing FCL and G-FCL mostly focuses only on the cognitive domain (Durrani et al., 2022; Lo & Hew, 2020), further studies are needed to examine students' affective domain in a STEM learning environment, considering variables such as grade, educational background, and gender. On the other hand, to enhance understanding of FCL in STEM education, more comparative studies with G-FCL are necessary. Thus, this research explores and examines the G-FCL model, as compared to the FCL model, which is based on blended and gamified approach (W. J. Baker, 2000; Kapp, 2012; Lage et al., 2000; Thongmak, 2019). The study aims to measure the effect of this methodology on students' affective domain (Holm et al., 2020; González-Gómez et al., 2022; Jeong & González-Gómez, 2022) for a university course, along with personal characteristics of pre-service teachers (PSTs) (Freeman et al., 2014; Jeong & González-Gómez, 2021). The research seeks to answer two research questions "Is the affective domain of students for learning contents greatly influenced by a G-FCL instruction methodology compared to FCL?" and "Do students' personal factors, such as grade, educational background, and gender, modulate the effects of the instruction methodology, regarding the affective domain?" The organization of this paper starts with the section "Methods and Materials," which describes the methods of research design and procedure, teachings methodologies, data collection, and quantitative data analysis. Then, the section "Results and Discussion" provides the results and discussion of the quantitative data analysis of affective domain and personal aspects of students. Finally, the section "Conclusions" concludes the research questions raised, implications and future research based on the limitations encountered.

Methods and materials

This study adopted a quasi-experimental design with quantitative data to explore and compare the FCL and G-FCL regarding students' affective domain, particularly their attitude and emotion, and their particular personal facets. The quasi-experimental design was adopted due to logistics limitation and to ensure that all participants benefit from the innovative instructional methods developed in this study.

Participants

The study involved four academic courses; it obtained the results into flipped classroom learning environments for a general mathematics subject. The FCL model was implemented during the 2019/20 and 2020/21 academic years, while the G-FCL model was used during the 2021/22 and 2022/23 academic years. A total of 242 PSTs participated in this quasi-experimental research through a quantitative online survey, which included informed consent approved by the university bioethics committee. As indicated in Table 1, there were 131 PSTs in the FCL group (average grade 7.56 and average age 19.98 years old) and 111 PSTs in the G-FCL group (average grade 7.39 and average age 20.42 years old). The FCL group included 76 female and 55 male PSTs, while the G-FCL group included 66 female and 45 male PSTs. Most PSTs came from a social science background and accessed the University primarily through high school degree. The demographics of the PSTs participating in this study were typical of similar contexts, with a predominance of females from a social science educational background.

Table 1. Demographic demonstrations of participants in the learning environments proposed.

Participant	Pre-grade (max. 10)	Age	Gender		Background			University access		
			Female	Male	Science	Social science	Technology	High school	Professional school	Test over age 25
FCL (131)										
64	7.59	20.00	41	23	24	33	7	60	3	1
67	7.50	19.95	35	32	17	48	2	59	6	2
G-FCL (111)										
59	7.47	20.31	36	23	22	32	5	56	3	0
52	7.31	20.52	30	22	12	36	4	46	5	1

Regardless of the learning environments, whether FCL or G-FCL, there was same professor and maintained the same contents and instructional methods based on the official syllabus specified by the University.

Research procedures

The research conducted during the 15 weeks, spanning one academic semester, consisted of 5 phases as it is shown in Figure 1. The course syllabus includes various types of numbers and their arithmetic operations, along with mathematical didactic methods that PSTs can use to teach their future elementary students.

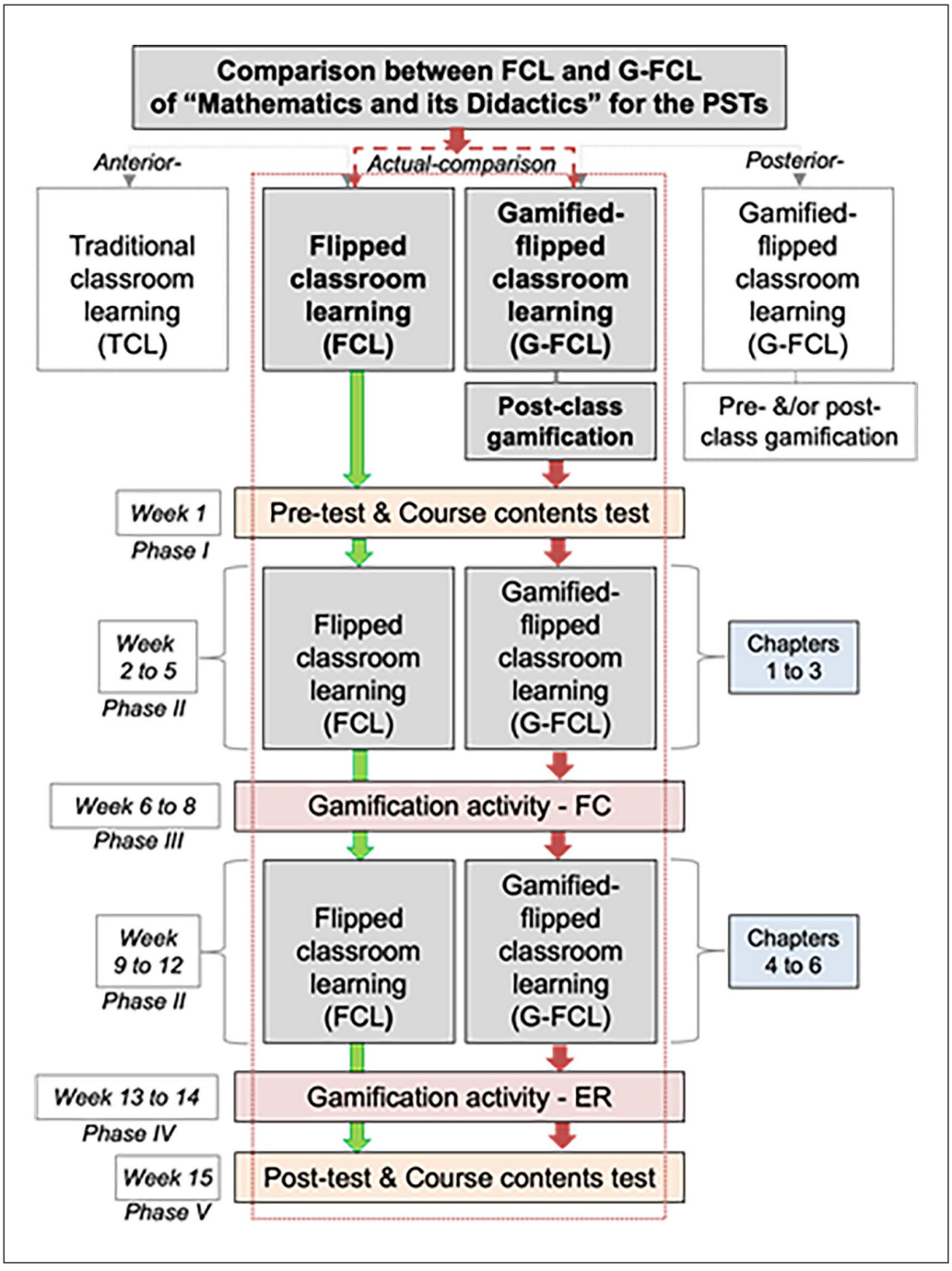

Figure 1. General research procedure to examine and compare the FCL and G-FCL for the impact on PSTs' affective domain and personal aspects of university mathematics course (FC [future classroom] and ER [escape room]).

The class was taught twice a week, totaling 4 hr. In the Phase I, during the first week, student completed an anonymous homogeneity test on the class contents, assessing their conceptual understanding and problem-solving abilities, and an anonymous questionnaire about their affective domain toward mathematics. In the Phase II, over 8 weeks, six chapters of syllabus were adopted into the flipped classroom model, as shown in Figure 2. In the Phases III and IV, students executed an intermediate project related to the chapters learnt, particularly chapters 1 to 3, for 3 weeks, and a final project related to all chapters learnt for 2 weeks. Finally, in the Phase V, during the last week, students completed an anonymous test on the class contents, assessing their conceptual understanding and problem-solving abilities, and an anonymous questionnaire about their affective domain toward mathematics, like the Phase 1.

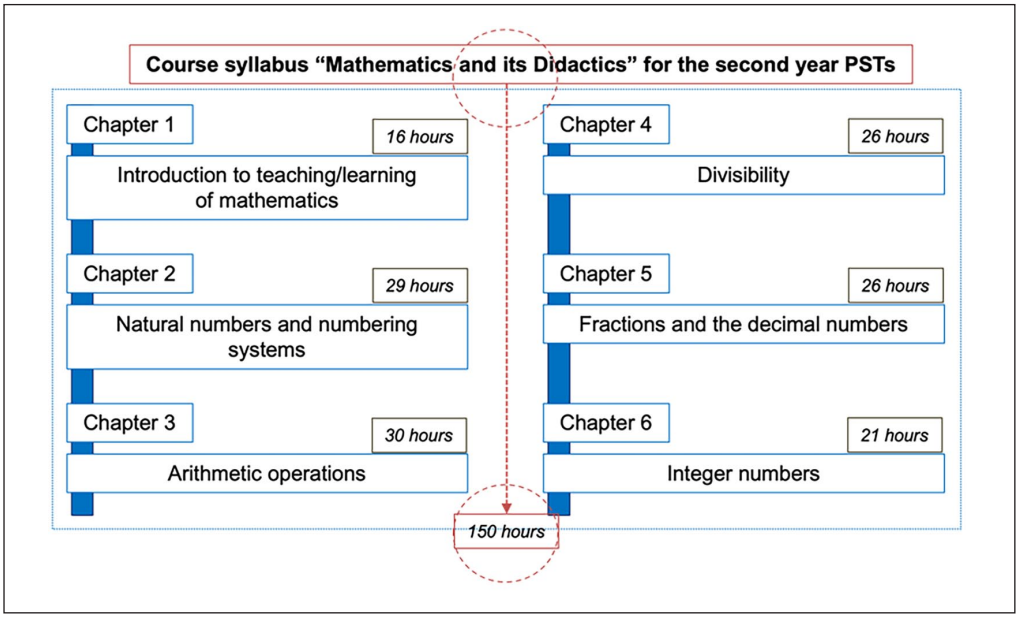

Figure 2. Course syllabus "Mathematics and its Didactics" for the second year PSTs.

Instruments as FCL and G-FCL environments and methodologies

Based on various previous research on flipped classroom, gamification, and their integration (Ekici, 2021; Hammill et al., 2021; Liao et al., 2019; Sun et al., 2021), Figure 3 shows the scheme of FCL and G-FCL instruments that contain various game design components created and utilized for this proposed research. In the Phase I, while students did not have any previous knowledge about the teaching methodology for the course, the flipped and gamification paradigm was introduced along with its scheduled flowchart. First, in the Phase II, through a Moodle virtual interface, as it is displayed in Figure 3a, students could access all the flipped classroom contents and materials available according to scheduled flowchart, such as written materials, video lessons, flash simulators, etc. All participants were enrolled in the Moodle platform before starting the course, and their progress was tracked using the "activity completion" feature in the Moodle platform. Additionally, there was a series of on-line quiz to review class contents. At the same time, students

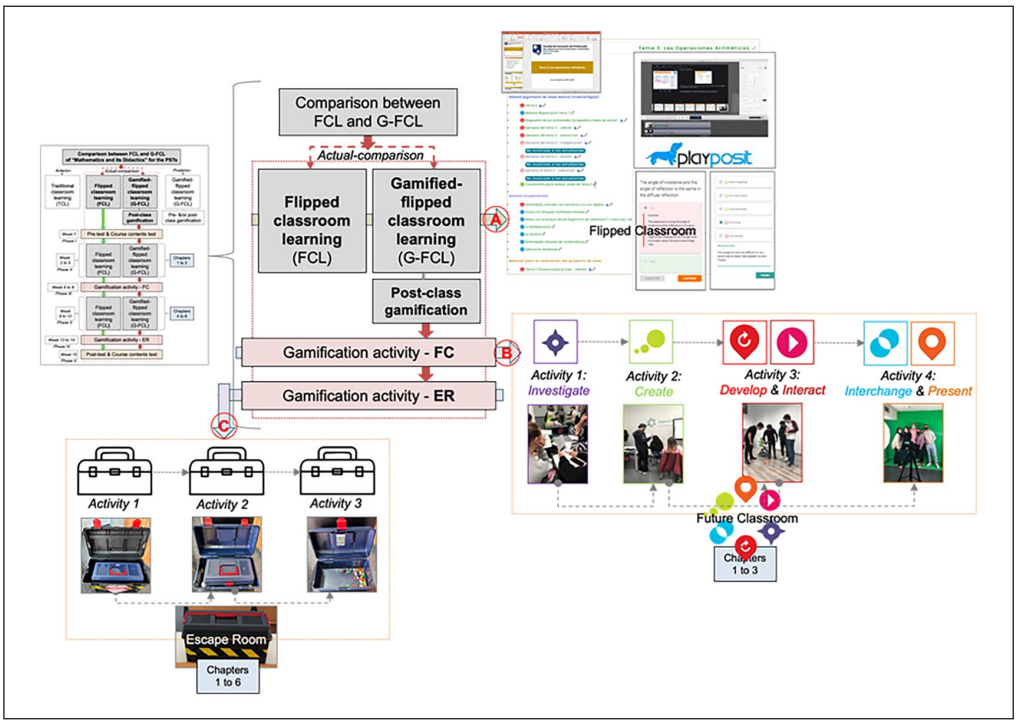

Figure 3. Scheme of FCL and G-FCL instruments: (a) flipped classroom learning, (b) future classroom (FC), and (c) escape room (ER).

could provide feedback on both the video lessons and the Moodle interface to the professor, who would deliver short "just-in-time" teaching in the classroom. Second, in the Phase III, through a future classroom (FC), students can revise all the contents of chapters 1 to 3 in six different zones: investigate, create, develop, interact, interchange, and present. Here, students can utilize these learning zones in the context of the objectives and necessities corresponding to the activity, which promote gamified collaborative learning and improve competence development as it is shown in Figure 3b. For example, in each zone with a small group of students, there were a gamified activity to learn numbers, numbering systems, and arithmetic operations in mathematics didactics along with new mobile and computer applications of slow motion (Life Lapse) and augmented reality (Quiver) together with digital and crystal board and chroma screen space in the FC. These activities were based on the necessity of learners, which allows the instructor to be turned to a facilitator and postulate feedback between them. Thus, the FC itself as a classroom promotes collaborative learnings and competence development of participants. Therefore, this approach fosters creativity, innovation, autonomy, and problem-solving of students (Ludeña, 2019). Finally, in the Phase IV, through an escape room (ER) as an educational tool, students can again revise all the theoretical contents of all chapters in a gamified manner. The mathematics ER implemented in this phase focused on numbers and arithmetic, exploring concepts such as the nature of the numbers, numerical systems, and arithmetic operations. The activities were designed to engage students in applying these concepts though a series of tasks. Briefly, the ER was contextualized with three different boxes, in a lineal sequence model, where students need to find proper codes as shown in Figure 3c. Thus, students need to find a way to open the different boxes by unlocking the padlocks codes.

These codes were obtained by completing six different tasks. The ER starts with a hidden code that students need to decipher by applying the lesson's contents. The following two tasks, different uncomplete tables are provided to the students that need to complete by applying various methods of number decomposition such as the sieve of Eratosthenes. The fourth task challenged students to solve a word search based on general topics covered in the lesson. The fifth activity required students to decipher a code by solving numerical problems. In all cases, points were given on a completion-time base. Ultimately, they can reach a final challenge (building a cubical manipulative object with specific dimensions based on the calculations given) to escape a classroom, interacting with the professor who can validate their performances. This method enhance collaboration, creativity, innovation, autonomy, and problem-solving in students (Ludeña, 2019).

During the activity of post-class gamification, PSTs will receive activity points as gaming elements that can promote a mutual challenge and animate students' learning consciousness (R. Baker et al., 2008; Hew et al., 2016).

Data collection and instrument

With the quantitative online-based pre- and post-test questionnaire, data were collected based on a 5-point Likert-type scale to measure the students' affective domain, toward mathematics learning. This questionnaire was adapted from previously published and validated research on the affective domain. For attitude, it was adapted from Auzmendi (1992) and Baroody and Coslick (1998) and, for emotion, the objectives proposed were adapted from Dunbar et al. (2016), Jeong et al. (2016) and Pekrun et al. (2007). Additionally, various university mathematics education professors, who were not related with this research, were requested to review and validate all the tests and questionnaires before the submission and approval of university bioethics committee as a pre-preparation process. Finally, the questionnaire and its protocols were approved by the bioethics committee of the authors' University (protocol numbers 94/2018 and 77/2023) to collect the PSTs' data before commencing the research. The questionnaire entailed of 39 questions divided into two sections related to mathematics learnings. The first section contained of 25 questions aimed at measuring the PSTs' attitudes on both in positive and negative scales. The second section contained 14 questions aimed at measuring the PSTs' emotions on both in positive and negative scale.

Data analysis

The Cronbach's α test was applied to check the instrument reliability, as a measure of the instrument internal consistency. The instrument consisted of two sections. For the first one, where the attitude toward mathematics learning were gauged, the Cronbach's α was 0.973. For the second part, dedicated to evaluating the emotion felt by the students when learning mathematics, the Cronbach's α was 0.911. Therefore, both sections of the instrument could be considered as reliable (Cho & Kim, 2015). Besides that, the reliability of the instrument was also assessed as a whole, in that case, the Cronbach's α was 0.976. As a result, in all cases, the Cronbach's α test can be determined as acceptable (Biggs et al., 2001). Based on the reliability results, two new variables were calculated. Firstly, a global students' attitude was computed as followed: $GA = \sum A_{positive} - \sum A_{negative}$. In the same way, a global emotion variable was calculated as $GE = \sum E_{positive} - \sum E_{negative}$.

Before assessing the influence of the instruction, on the studied variables, sample homogeneity was tested by comparing participants' grades before taking the course (pre-test grade). To do this, the Kolmogorov–Smirnov test for normality was used to check whether pre-test grade data were normally distributed or not. Because this data was normally distributed ($p = .865$), the homogeneity

of both groups was assessed by testing the existence of significant differences between the mean values of the pre-test grade by the Student's *t test*. The mean pre-test grade for the FCL group was 7.54 (std dev = .84), whereas the mean pre-test grade for the G-FCL group was 7.39 (std dev = .83). A Student's independent sample *t-test* showed that difference between them was not significant (t(240) = 1.40), $p = .620$), suggesting that both groups were homogeneous in terms of their previous grade before starting the course to be evaluated. In addition, the homogeneity was also tested in terms of students' affective domain before starting the course. For that purpose, a pooled affective variable was measured as $GA_f = GA + GE$. Thus, the mean values of the pre-test GA_f were compared between groups by means of the Students' *t test* (Kolmogorov–Smirnov test $p = .891$), indicating that no significant difference was observed between both values (t(240) = 1.56), $p = .121$), being the mean values for this variable 27.2 ± 1.89 and 22.9 ± 2.02 for the FCL and G-FCL, respectively. Therefore, the sample was homogeneous in terms of pre-test grade and affective domain.

To test the influence of the instructional methodologies (FCL *vs.* G-FCL) on the studied variables, an analysis of variance (ANOVA) was conducted. When significant differences were found, the effect size was measured (η^2). Additionally, a post-hoc test was performed to identify significant differences between groups. When the post-hoc analysis revealed the existence of significant differences, the effect size was measured using Cohen's *d*. Finally, to assess any possible interaction between participants' educational background and/or gender on the studied variables related to mathematics learning, a factorial ANOVA test was applied. When significant differences were observed, the effect size was additionally measured (η^2). Furthermore, to evaluate the influence of students' grade on the dependent variables after the instruction, an analysis of covariance (ANCOVA) test was conducted. Finally, a linear regression analysis was performed to examine the relationships between the different variables involved in this study.

Results and discussion

Aiming to answer the proposed research questions and, once the sample homogeneity was verified, both in terms of pre-grade and affective domain, a comparative assessment of the studied variables was conducted to gauge the influence of the instruction methodology (FCL *vs*. G-FCL). Particularly, the attitude toward mathematics learning were first compared before and after the intervention in both groups. Later, similar study was conducted for the emotion. In both cases, a significant influence was detected. Figure 4 summarizes the results obtained after conducting the comparative assessment analysis.

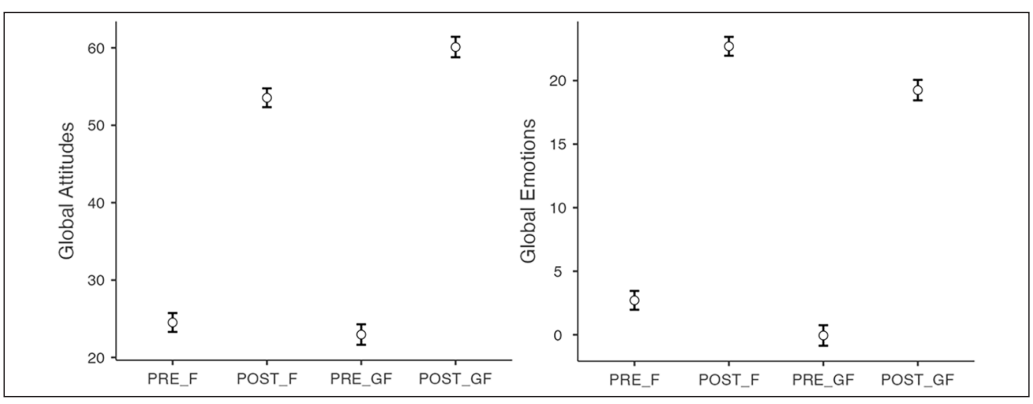

Figure 4. Mean values for the Global Attitudes (left) and Global Emotions (right) felt by the students before and after the intervention (FCL *vs.* G-FCL, pre- and post-test).

Influence of the instruction methodology on the students' attitude

The influence of the instruction methodology on the students' attitude was assessed by conducting an ANOVA test (pre- and post-intervention measurements *vs.* FCL and G-FCL). Particularly, the ANOVA test revealed a significant effect on the students' global attitudes (GA) based on the intervention ($F(3, 480) = 228$, $p < .001$). The effect size, measured as eta-squared, was .594 indicating that the instruction methodology had a large influence on the variability of the students' attitudes toward mathematics learning. To have a better understanding of the influence of each methodology (FCL and G-FCL) a post hoc test, using the Holm correction to adjust p, was used. The test indicated that students' attitudes were significant different depending on the methodology ($p = .001$). Precisely, the GA mean value after the intervention was 60.1 (SE = 1.3) for the G-FCL methodology, that was significantly higher than the value received for the FCL methodology (53.6; SE = 1.2) as it is summarized in Table 2. When analyzing the Cohen's d effect size, it is noticeable the large effect that the intervention has in the students' global attitudes (pre- and post-results in both methodologies), where this indicator shows a very large effect size. Finally, when the effect of the students' grade was considered in the study, the ANCOVA test suggested that the "pre-grade" F value (was significant at $p = .019$, suggesting that this covariable is a relevant predictor of the dependent variable, global attitudes.

Table 2. Post hoc comparisons (Holm correction) for the "global attitudes" variable before and after the instruction (FCL *vs.* G-FCL, Pre- and Post-test).

Attitudes comparison	Mean difference	*SE*	*df*	*t*	p_{holm}	Cohen's *d*	95% confidence interval [lower, upper]
Pre-FCL							
Post-FCL	−29.05	1.72	480	−16.869	< .001	−2.084	[−2.361, −1.808]
Pre-G-FCL	1.56	1.80	480	0.866	.387	0.112	[−0.142, 0.365]
Post-G-FCL	−35.60	1.80	480	−19.800	< .001	−2.554	[−2.855, −2.254]
Post-FCL							
Pre-G-FCL	30.60	1.80	480	17.022	< .001	2.196	[1.907, 2.485]
Post-G-FCL	−6.55	1.80	480	−3.644	< .001	−0.470	[−0.725, −0.215]
Pre-G-FCL							
Post-G-FCL	−37.15	1.87	480	−19.862	< .001	−2.666	[−2.979, −2.353]

Influence of the instruction methodology on the students' emotion

Following a similar procedure as for the previous variable of attitude, the students' emotion was assessed by means of the ANOVA test. Again, the ANOVA test indicated that there was a significant influence of the instruction methodology on the students' emotion toward mathematics learning ($F(3, 480) = 858$, $p < .001$). For this variable, the effect size measured as eta-squared was .843, denotating a very large effect of the instruction methodology on the students' emotion. When a post hoc test (Holm correction to adjust p) was conducted, the influence of each individual methodology (FCL and G-FCL before and after the intervention) was elucidated. For this case, and as it is shown in Figure 4, the mean values for this variable are statistically higher for the FCL group compared to G-FCL group as it is summarized in Table 3, however the mean difference in both cases are similar, 19.98 (SE = 0.53) for the pre- and post-test for the FCL and 19.31 (SE = .58) for

Table 3. Post hoc comparisons (Holm correction) for the "global emotions" variable before and after the instruction (FCL *vs.* G-FCL, Pre- and Post-test).

Emotions comparison	Mean difference	*SE*	*df*	*t*	p_{holm}	Cohen's *d*	95% confidence interval [lower, lower]
Pre-FCL							
Post-FCL	−19.98	0.534	480	−37.46	<.001	−4.628	[−5.009, −4.247]
Pre-G-FCL	2.76	0.557	480	4.96	<.001	0.640	[0.384, 0.897]
Post-G-FCL	−16.54	0.557	480	−29.70	<.001	−3.831	[−4.182, −3.480]
Post-FCL							
Pre-G-FCL	22.75	0.557	480	40.84	<.001	5.268	[4.850, 5.687]
Post-G-FCL	3.44	0.557	480	6.18	<.001	0.797	[0.797, 1.056]
Pre-G-FCL							
Post-G-FCL	−19.31	0.580	480	−33.31	<.001	−4.471	[−4.471, −4.085]

the pre- and post-test for the G-FCL instruction methodology. Thus, in both cases (FCL and G-FCL), the methodology had a significant influence on the students' emotion, significantly increasing the mean values in both cases, and this increase was of the same magnitude in both groups. For this variable, the Cohen's *d* effect size measurement indicates a very large effect of the instruction methodology on the students' global emotions. In addition, when the students' grade was considered as covariate, a significant effect at $p = .001$ was obtained after completing the ANCOVA test, suggesting that the students' pre-grade has a significant predictor in the students' global emotions.

Multiple regression analysis considering the instruction methodology and students' pre-grade, educational background, and gender

Once a significant effect of the instruction methodology on the dependent variables was established, a regression study was conducted to assess the relationship between the different factors involved in the study. Specifically, a linear regression analysis was conducted for the global attitudes and global emotions' variables. In this analysis, students' pre-grade values were used as covariates, and the pre- and post-instruction moments were considered as factors. Additionally, influence of the students' educational background and gender was also considered. In Table 4, the linear regression model is summarized.

According to the results for the global attitudes, the linear regression model was statistically significant ($F(7, 476) = 178$, $p < .001$), and explained 69% of the variance in the global attitude ($R^2 = .69$). The model coefficients are summarized in Table 4, and it is shown the students' grade before starting the study, the instruction methodology (FCL and G-FCL before and after the intervention) and the educational background were significant predictors for the students' global attitudes as it is indicated in Table 4. On the other hand, regarding the results for the global emotions, the lineal model was statistically significant as well ($F(7, 476) = 618$, $p < .001$). In this case, the model explained 89% of the variance in the global emotions ($R^2 = .89$). The model coefficients are displayed in Table 4 showing that the students' grade before starting the study, the instruction methodology (FCL and G-FCL before and after the intervention) as well as the students' background were significant predictors for the students' global emotions. As it is displayed in Table 4, the gender was not a significant predictor neither for the students' global attitudes nor students' global emotions.

Table 4. Multiple regression model and standardized coefficients for the global attitudes and global emotions felt by the students during the study.

Linear regression model	Coefficients global attitudes				Coefficients global emotions			
	R	R^2	*F*	*p*	*R*	R^2	*F*	*p*
	.831	.691	178	<.001	.941	.886	618	<.001
Predictor	Estimate	*SE*	*T*	*p*	Estimate	*SE*	*t*	*p*
(Intercept)	21.497	5.191	4.141	<.001	7.282	1.534	4.748	<.001
Pre-grade	1.533	0.663	2.314	.021	0.679	0.202	3.361	<.001
FCL vs. G-FCL								
Post-FCL – Pre-FCL	29.046	1.495	19.431	<.001	19.985	0.456	43.848	<.001
Pre-G-FCL – Pre-FCL	−1.281	1.564	−0.819	.413	−2.645	0.477	−5.548	<.001
Post-G-FCL – Pre-FCL	35.872	1.564	22.938	<.001	16.661	0.477	34.94	<.001
Educational background								
Social science – science	−13.89	1.210	−11.484	<.001	−4.522	0.369	−12.25	<.001
Technology – science	0.554	2.298	0.241	.810	−0.350	0.701	−0.500	.617
Gender								
Male-female	1.877	1.155	1.625	.105	0.274	0.353	0.776	.438

Discussion of the results

The role of the affective domain in the learning process has been proved to be an essential factor to ensure students to achieve better learning outcomes (Bandura, 1971; Jeong & González-Gómez, 2021; McLeod, 1989; Pekrun et al., 2007). To address learning difficulties, it is important to design instruction methodologies that appeals not only to the cognitive aspects of the learning process but also the affective domain. This is even more relevant for subjects where students tend to struggle, such as in mathematics (Jeong & González-Gómez, 2022), and when the students' educational background is not solid enough (Holtmann et al., 2024). In these cases, it has been proved that the instruction methodology should effectively promote students' affective domain to ensure a significative learning (Bellocchi et al, 2013; Joensuu-Salo et al., 2024; Kyriaki Kourti & Potari, 2024; Wadaani, 2023). Different research studies have proved that the flipped classroom, as an active learning, applied to mathematics could be a successful approach (Mattis, 2015). However, this methodology must overcome the rapid detachment that students feel toward learning this content (Bhagat et al., 2016; Clark, 2015; Lo & Hew, 2020). In a flipped classroom methodological environment, this translates, as pointed out by Sahin et al. (2015), into less student involvement in the learning process and in the autonomous preparation of content prior to their arrival in the classroom, as corresponds in a flipped instruction model. In this sense, and with the aim of alleviating this situation, different alternatives have been proposed. One of them consists of integrating gamification strategies into the flipped classroom methodology (Braun et al., 2014; Lo & Hew, 2020; Parra-González et al., 2021; Xiao & Hew, 2024). The inclusion of gamification in the FLC has shown to increase significantly the students' attitudes toward mathematics learning compared to the situation in which no gamification was included. Although in both settings (FCL and G-FCL) a significant influence was observed, it was higher in the G-FCL. With respect to the students' emotion, in both settings, a significant increase is observed after the instruction. In this case, there is not significant differences between the FLC and G-FLC (post-test), but it is relevant how in both cases a very significant effect is observed (see Cohen's *d* value in Table 3). It has been described that one of the main causes of poor learning outcomes, especially in science and mathematics, is due to emotional factors (Solbes, 2011; Solbes &

Tarín, 1998). Thus, ignoring the affective aspects of learning can limit the student's conceptual change (Duit & Treagust, 2008). Consequently, teaching strategies and instructional methods should encourage positive emotional states, as these promote learning and foster active learning in the classroom, while negative emotional states limit it (Aydogan et al., 2015; Jarvis & Pell, 2004; Osborne et al., 2003). Therefore, the instructional method followed must contribute to controlling and regulating the affective domain of the student involved in the teaching and learning process.

When assessing the influence of the instruction methodology in the affective domain, it is also revealed to consider other students' personal factors that could modulate their feelings toward the learning process (Faber et al., 2017; González-Gómez et al., 2020; Holtmann et al., 2024; Jeong & González-Gómez, 2021). In that sense, the research indicated that high performance group had more significant results in affective domain through active learning methodology toward mathematics education. Also, it confirms that students not having a solid background in mathematics would benefit from a teaching methodology in which their affective domain is fostered, like the case of FCL and more specifically the case of G-FCL as it is shown in Table 4. That is especially relevant in the context of the PSTs' education, where the number of the students without solid mathematics contents is higher than in other learning context. In that sense, Bego et al. (2020) indicated in their study that when a flipped methodology is applied in a university course, contributed to reduce the number of dropouts and increased the success rate in a mathematics course. Besides that, Maarif et al. (2022) reported that a FCL environment improved the learning capacity of the students, and that would be enhance when this methodology is complemented with gamification. Thus, Jeong and González-Gómez (2021) confirmed that there was no difference of affective domain in gender toward mathematics learning neither in flipped classroom nor gamified-flipped classroom.

Conclusions

Active learning methodologies have been proved to be effective to achieve significant learning in PSTs. The application of such methodologies in different educational context, not only contribute to attain better learning outcomes (cognitive domain) but also foster more positive levels in the affective domain, which also have a positive impact in the learning process. In this context, the flipped classroom learning has successfully fulfilled these premises, being an appropriate active methodology for higher mathematics education. However, there is a need to prevent the students' detachment of this learning context, being the gamification as a promising strategy. Thus, this study has proved that coupling gamifications activities within the flipped classroom methodology (G-FCL) significantly increased the students' affective domain toward mathematics learning in a more significative level that non gamified flipped classroom does (FCL). This promotion is also affected by other personal aspects of students. Precisely, the effects on the affective domain were affected by the students' pre-grade (before the instruction) and their educational background, while participants gender did not have a significant influence. The proposed instruction methodology is not only feasible for PSTs' programs but also could be easily applied to other educational levels, adapting the contents and maintaining the suggested structure of the course setting. Thus, this study concludes that employing blended- and gamified-learning methodologies will have further benefits in the flipped classroom as an active learning based on the limitations encountered.

Limitations

The intrinsic nature of this study as a quasi-experimental study must be considered when interpreting the results. Although the study was conducted over 4 years and involved a relevant sample size, the results are constrained to this specific context in which the study was conducted.

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References

Alt, D. (2023). Assessing the benefits of gamification in mathematics for student gameful experience and gaming motivation. *Computers & Education*, *200*, Article 104806.

Aragón, E., Menacho, I., Navarro, J. I., & Aguilar, M. (2024). Teaching strategies, cognitive factors and mathematics. *Heliyon*, *10*(9), e29831.

Ashkenazi, S., & Velner, H. (2023). The interplay between math performances, spatial abilities, and affective factors: The role of task. *Trends in Neuroscience and Education*, *33*, Article 100211.

Auzmendi, E. (1992). *Las actitudes hacia la matemática-estadística en las enseñanzas medias y universitarias* [Attitudes towards mathematics-statistics in secondary and university education]. Mensajero.

Aydogan, H., Bozkurt, F., & Coskun, H. (2015). An assessment of brain electrical activities of students toward teacher's specific emotions. *International Journal of Social, Behaviors, Educational, Economics, Business and Industrial Engineering*, *9*(6), 1977–2000.

Baker, R., Walonoski, J., Heffernan, N., Roll, I., Corbett, A., & Koedinger, K. (2008). Why students engage in "gaming the system" behavior in interactive learning environments. *Journal of Interactive Learning Research*, *19*(2), 185–224.

Baker, W. J. (2000). *The "classroom flip": Using web course management tools to become the guide by the side*. Cedarville University: Communication Faculty Publication.

Bandura, A. (1971). *Social learning theory*. General Learning Press.

Baroody, A. J., & Coslick, R. T. (1998). *Fostering children's mathematical power. An investigative approach to K-8 mathematics instruction*. Lawrence Erlbaum Ass.

Bassett, K., Olbricht, G. R., & Shannon, K. B. (2020). Student pre-class preparation by both reading the textbook and watching videos online improves exam performance in a partially flipped course. *CBE—Life Sciences Education*, *19*(3), 1–9.

Bego, C. R., Ralston, P. A. S., & Thompson, A. K. (2020). Improving performance in a large-flipped barrier mathematics course: A longitudinal case study. *International Journal of Mathematical Education in Science and Technology*, *53*(7), 1916–1933.

Bellocchi, A., Ritchie, S. M., Tobin, K., Sandhu, M., & Sandhu, S. (2013). Exploring emotional climate in preservice science teacher education. *Cultural Studies of Science Education*, *8*, 529–552.

Bhagat, K. K., Chang, C. N., & Chang, C. Y. (2016). The impact of the flipped classroom on mathematics concept learning in high school. *Educational Technology & Society*, *19*(3), 134–142.

Biggs, J. B., Kember, D., & dan Leung, D. Y. P. (2001). The revised two-factor study process questionnaire: R-SPQ-2F. *British Journal of Educational Psychology*, *71*, 133–149.

Bishop, J. L., & Verleger, M. (2013, June 23). *The flipped classroom: A survey of the research* [Paper presentation]. 120th ASEE Annual Conference and Exposition, Atlanta, GA.

Blanco, L. J., Guerrero, E., Caballero, A., Brígido, M., & Mellado, V. (2010). The affective dimension of learning and teaching mathematics and science. In M. P. Caltone (Ed.), *Handbook of lifelong learning developments* (pp. 265–287). Springer.

Bonwell, C. C., & Eison, J. A. (1991). *Active learning: Creating excitement in the classroom* (ASHE-ERIC Higher Education Report No. 1). School of Education and Human Development. The George Washington University.

Bower, G. H. (1981). Mood and memory. *American Psychologist*, *36*, 129–148.

Braun, I., Ritter, S., & Vasko, M. (2014). Inverted classroom by topic: A study in mathematics for electrical engineering students. *International Journal of Engineering Pedagogy*, *4*(3), 11–17.

Chang, S. J., Kim, G. M., & Kim, J. A. (2024). The effects of flipped learning and gamification on nursing students' patient safety education: A mixed method study. *Heliyon*, *10*(8), e29538.

Chen, C.-H., & Law, V. (2016). Scaffolding individual and collaborative game-based learning in learning performance and intrinsic motivation. *Computers in Human Behavior*, *55*, 1201–1212.

Cho, E., & Kim, S. (2015). Cronbach's coefficient alpha: Well known but poorly understood. *Organizational Research Methods*, *18*(2), 207–230.

Clark, K. R. (2015). The effects of the flipped model of instruction on student engagement and performance in the secondary mathematics classroom. *Journal of Educators Online*, *12*(1), 91–115.

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011, September 28–30). *From game design elements to gamefulness: Defining gamification* [Conference session]. 15th International Academic MindTrek Conference: Envisioning Future Media Environments, Tampere, Finland.

Dichev, C., & Dicheva, D. (2017). Gamifying education: What is known, what is believed and what remains uncertain: A critical review. *International Journal of Educational Technology in Higher Education*, *14*(1), 1–36.

Dicheva, D., Dichev, C., Arge, G., & Angelova, G. (2015). Gamification in education: A systematic mapping study. *Educational Technology & Society*, *18*(3), 75–88.

Duit, R., & Treagust, D. W. (2008). Teaching science for conceptual change: Theory and practice. In S. Vosniadou (Ed.), *Metacognition: New research developments* (pp. 137–151). Nova Science Publisher.

Dunbar, R. L., Dingel, M. J., Dame, L. F., Winchip, J., & Petzold, A. L. (2016). Student social self-efficacy, leadership status, and academic performance in collaborative learning. *Studies in Higher Education*, *43*(9), 1507–1523.

Durrani, U. K., Naymat, G. A., Ayoubi, R. M., Kamal, M. M., & Hussain, H. (2022). Gamified flipped classroom versus traditional classroom learning: Which approach is more efficient in business education? *The International Journal of Management Education*, *20*, Article 100595.

Ekici, M. (2021). A systematic review of the use of gamification in flipped learning. *Education and Information Technologies*, *26*, 3327–3346.

Faber, J. M., Luyten, H., & Visscher, A. J. (2017). The effects of a digital formative assessment tool on mathematics achievement and student motivation: Results of a randomized experiment. *Computers & Education*, *106*, 83–96.

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of evidence. *Review of Education Research*, *74*(1), 59–109.

Freeman, S., Eddy, S. L., McDonugh, M., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Psychological and Cognitive Science*, *111*(23), 8410–8415.

Furinghetti, F., & Morselli, F. (2009). Every unsuccessful problem solver is unsuccessful in his or her own way: Affective and cognitive factors in proving. *Educational Studies in Mathematics*, *70*. 71–90.

González-Gómez, D., Jeong, J. S., & Cañada-Cañada, F. (2020). Examining the effect of an online formative assessment tool (OFAT) of students' motivation and achievement for a university science education. *Journal of Baltic Science Education*, *19*(3), 401–414.

González-Gómez, D., Jeong, J. S., & Cañada-Cañada, F. (2022). Enhancing science self-efficacy and attitudes of Pre-Service Teachers (PST) through a flipped classroom learning environment. *Interactive Learning Environments*, *30*(5), 896–907.

Gündüz, A. Y., & Akkoyulu, B. (2020). Effectiveness of Gamification in Flipped Learning. *SAGE Open*, *10*(4), 1–16.

Hammill, J., Nguyen, T., & Henderson, F. (2021). Encouraging the flip with a gamified process. *International Journal of Educational Research Open*, *2*, Article 100085.

Hao, Y. (2016). Exploring undergraduates' perspectives and flipped learning readiness in their flipped classrooms. *Computers in Human Behavior*, *59*, 82–92.

Hascher, T., & Edlinger, H. (2009). Positive emotions and well-being in school: An overview of methods and results. *Psychologie in Erziehung und Unterricht*, *56*, 105–122.

Hew, K. F., Huang, B., Chu, K. W. S., & Chiu, D. K. (2016). Engaging Asian students through game mechanics: Findings from two experiment studies. *Computers & Education*, *92*, 221–236.

Holm, M. E., Björn, P. M., Laine, A., Korhonen, J., & Sakari Hannula, M. (2020). Achievement emotions among adolescents receiving special education support in mathematics. *Learning and Individual Differences*, *79*, Article 101851.

Holtmann, M., Rjosk, C., Jansen, M., & Lüdtke, O. (2024). The double-edged sword of de-tracking reforms in Germany: Using a Difference-in-Differences approach to estimate effects on students' educational aspirations and their association with students' socioeconomic backgrounds. *Studies in Educational Evaluation*, *81*, Article 101350.

Huang, B., & Hew., K. F. (2018). Implementing a theory-driven gamification model in higher education flipped courses: Effects on out-of-class activity completion and quality of artifacts. *Computers & Education*, *125*, 254–272.

Huang, B., Hew, K. F., & Lo, C. K. (2019). Investigating the effects of gamification-enhanced flipped learning on undergraduate students' behavioral and cognitive engagement. *Interactive Learning Environments*, *27*(8), 1106–1126.

Jarvis, T., & Pell, A. (2004). Primary teachers' changing attitudes and cognition during a two-year science in-service program and their effect on pupils. *International Journal of Science Education*, *26*, 1787–1811.

Jeong, J. S., & González-Gómez, D. (2021). A STEM course analysis during COVID-19: A comparison study in performance and affective domain of PSTs between F2F and F2S flipped classroom. *Frontiers in Psychology*, *12*, Article 669855.

Jeong, J. S., & González-Gómez, D. (2022). Mathematics self-belief comparison and examination of pre-service teacher (PST) through a flipped-open calculation based on numbers (ABN) learning method. *Heliyon*, *8*(7), e09806.

Jeong, J. S., González-Gómez, D., & Cañada-Cañada (2016). Students' perceptions and emotions toward learning in a flipped general science classroom. *Journal of Science Education and Technology*, *25*, 747–758.

Joensuu-Salo, S., Hämäläinen, M., Peltonen, K., & Raappan, A. (2024). The temporal development of teachers' attitudes on entrepreneurship education: A latent growth curve analysis. *Teaching and Teacher Education*, *144*, Article 104600.

Kapp, K. M. (2012). *The gamification of learning and instruction: Game-based methods and strategies for training and education*. John Wiley & Sons.

Kim, M. K., Kim, S. M., Khera, O., & Getman, J. (2014). The experience of three flipped classrooms in an urban university: An exploration of design principles. *The Internet and Higher Education*, *22*, 37–50.

Kyriaki Kourti, S., & Potari, D. (2024). Teacher emotions and in-the-moment decision making in the secondary mathematics classroom. *The Journal of Mathematical Behavior*, *74*, Article 101152.

Lage, M. J., Platt, G. J., & Treglia, M. (2000). Inverting the classroom: A gateway to creating an inclusive learning environment. *The Journal of Economic Education*, *31*(1), 30–43.

Lee, Y. -J. (2011). A study on the effect of teaching innovation on learning effectiveness with learning satisfaction as a mediator. *World Transactions on Engineering and Technology Education*, *9*(2), 92–101.

Li, N., Chee Mok, I. A., & Cao, Y. (2019). The evolution of mathematical thinking in Chinese mathematics education. *Mathematics*, *7*, 297.

Liao, C. -W., Chen, C. -H., & Shih, S. -J. (2019). The interactivity of video and collaboration for learning achievement, intrinsic motivation, cognitive load, and behavior patterns in a digital game-based learning environment. *Computers & Education*, *133*, 43–55.

Lo, C. K., & Hew, K. F. (2020). A comparison of flipped learning with gamification, traditional learning, and online independent study: The effects on students' mathematics achievement and cognitive engagement. *Interactive Learning Environments*, *28*(4), 464–481.

Lo, C. K., Hew, K. F., & Chen, G. (2017). Toward a set of design principles for mathematics flipped classrooms: A synthesis of research in mathematics education. *Educational Research Review*, *22*, 50–73.

Ludeña, E. S. (2019). La educación STEAM y la cultura maker (STEAM education and culture maker). *Journal of Parents and Teachers*, *379*, 45–51.

Maarif, S., Umam, K., Soebagyo, J., & Pradipta, T. R. (2022). Critical review on mathematics virtual classroom practice in private university. *International Journal of Nonlinear Analysis and Applications*, *13*(1), 975–982.

Mattis, K. V. (2015). Flipped classroom versus traditional textbook instruction: Assessing accuracy and mental effort at different levels of mathematical complexity. *Technology, Knowledge and Learning*, *20*(2), 231–248.

McLeod, D. B. (1989). Beliefs, attitudes, and emotions: New view of affect in mathematics education. In D. B. McLeod & V. M. Adams (Eds.), *Affect and mathematical problem solving: A new perspective* (pp. 245–258). Springer-Verlag.

Michael, J. (2006). Where's the evidence that active learning works? *American Journal of Physiology - Advances in Physiology Education*, *30*(4), 159–167.

Moraros, J., Islam, A., Yu, S., Banow, R., & Schindelka, B. (2015). Flipping for success: Evaluating the effectiveness of a novel teaching approach in a graduate level setting. *BMC Medical Education*, *15*, 1–10.

Nederveld, A., & Berge, Z. L. (2015). Flipped learning in the workplace. *Journal of Workplace Learning*, *27*(2), 162–172.

Nicholson, S. (2012, June). *A user-centered theoretical framework for meaningful gamification* [Paper presentation]. Games+Learning+society 8.0, Madison, WI.

O'Flaherty, J., & Phillips, C. (2015). The use of flipped classrooms in higher education: A scoping review. *Internet and Higher Education*, *25*, 85–95.

Osborne, J., Simon, S., & Collins, S. (2003). Attitudes towards science: A review of the literature and its implications. *Instrumental Journal of Science Education*, *25*, 1049–1079.

Parr-González, M. E., López-Belmonte, J. L., Segura Robles, A., & Moreno Guerrero, A. J. (2021). Gamification and flipped learning and their influence on aspects related to the teaching-learning process. *Heliyon*, *7*(2), e06254.

Pekrun, R., Frenzel, A. C., Goetz, T., & Perry, R. P. (2007). The control-value theory of achievement emotions: An integrative approach to emotions in education.In P. A. Schutz & R. Pekrun (Eds.), *Emotion in education* (pp. 13–36). Elsevier Academic Press.

Pho, A., & Dinscore, A. (2015). *Game-based learning*. Tips and trends.

Pintrich, P. R., Marx, R. W., & Boyle, R. A. (1993). Beyond cold conceptual change: The role of motivational beliefs and classroom contextual factors in the process of conceptual change. *Review of Educational Research*, *63*, 167–199.

Prensky, M. (2003). Digital game-based learning. *Computer Entertainment*, *1*(1), 21.

Prince, M. J. (2004). Does active learning work? A review of the research. *Journal of Engineering Education*, *93*(3), 223–231.

Roach, T. (2014). Student perceptions toward flipped learning: New methods to increase interaction and active learning in economics. *International Review of Economics Education*, *17*, 74–84.

Roberts, D. (2017). Higher education lectures: From passive to active learning via imagery? *Active Learning in Higher Education*, *20*(1), 63–77.

Sahin, A., Cavalzoglu, B., & Zeytuncu, Y. E. (2015). Flipping a college calculus course: A case study. *Educational Technology & Society*, *18*(3), 142–152.

Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Computers in Human Behavior*, *69*, 371–380.

Schwarz, N. (1990). Feeling as information. informational and motivational functions of affective states. In E. T. Higgins & R. Sorrentino (Eds.), *Handbook of motivation and cognition. Foundations of social behavior* (pp. 527–561). Guilford Press.

Solbes, J. (2011). ¿Por qué disminuye el alumnado de ciencias? [Why is the number of science students decreasing?] *Alambique*, *67*, 53–61.

Solbes, J., & Tarín, F. (1998). Algunas dificultades en torno a la conservación de la energía [Some difficulties around energy conservation]. *Enseñanza de las Ciencias*, *16*(3), 387–397.

Sun, L., Ruokamo, H., Siklander, P., Li, B., & Devlin, K. (2021). Primary school students' perceptions of scaffolding in digital game-based learning in mathematics. *Learning, Culture and Social Interaction*, *28*, Article 100457.

Thongmak, M. (2019). The student experience of student-centered learning methods: Comparing gamification and flipped classroom. *Education for Information*, *35*(2), 99–127.

Wadaani, M. (2023). The influence of preservice education and professional development in mathematics teachers' attitudes toward nurturing creativity and supporting the gifted. *Journal of Creativity*, *33*(1), Article 100043.

Williams, C., Perlis, S., Gaughan, J., & Phadtare, S. (2018). Creation and implementation of a flipped jigsaw activity to stimulate interest in biochemistry among medical students. *Biochemistry and Molecular Biology Education*, *46*, 343–353.

Williams, M., Uhing, K., Bennett, A., Voigt, M., Funk, R., Smith, W. M., & Donsig, A. (2022). Conceptualizations of active learning in departments engaged in instructional change efforts. *Active Learning in Higher Education*, *25*(3), 355–374.

Xiao, Y., & Hew, K. F. (2024). Personalized gamification versus one-size-fits-all gamification in fully online learning: Effects on student motivational, behavioral and cognitive outcomes. *Learning and Individual Differences*, *113*, Article 1024470.

Yoon, S., Kim, S., & Kang, M. (2020). Predictive power of grit, professor support for autonomy and learning engagement on perceived achievement within the context of a flipped classroom. *Active Learning in Higher Education*, *21*(3), 233–247.

Zichermann, G., & Cunningham, C. (2011). *Gamification by design: Implementing game mechanics in web and mobile apps*. O'Reilly Media Inc.

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